

Zak G Rackham

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Personal Profile

Computer science graduate and industry professional with experience in automation, robotics and digital imaging in a consultative capacity. Seeking a software development or deployment position within a systems or full stack orientated role to further develop technology or applications thereof to positively contribute towards a company's success and innovation.

Education

BSc (Hons) in Computer Science, 1st Class, University of Portsmouth (2022)

- Modules include Database design and development; Programming principles, modalities and applications; Web foundations and web markup; Network fundamentals; Data structures and algorithms; Discrete Mathematics and functional programming; Theoretical computer science and computability; Computer architecture; 3D computer graphics and animation; Operating systems and internetworking; Software engineering theory and practice; Distributed systems and security; Internet of things and Robotics.

College - Sept 2016 - July 2019 (Relevant)

- **A-Levels:** Business(C)
- **BTEC:** Applied Law (Distinction), Information Technology (Distinction*)

School - Sept 2011 - June 2016 (Relevant)

- **GCSE's:** 7 obtained at grade A-C, Mathematics (B), English (B) and Science (A)

Relevant Experience

Field Consultant, Oct 2018 – Aug 2019

Imagic Imaging Ltd (Imagic Bildverarbeitung AG)

- End-to-end preparation, installation, commissioning, and configuration of industrial digital image management systems across both server and client environments. This includes workstation setup, application server deployment, and integration of cameras, microscopes, and peripheral hardware.
- Conduct workflow analysis and design database architectures to support scalable, efficient operations. Implement LDAP integration and develop/consume RESTful APIs for interoperability.
- Provide ongoing system support, user training, and maintenance. Produce comprehensive reports and documentation covering full system implementation and lifecycle.

- Develop and maintain internal applications and customer data aggregation pipelines using Python, JavaScript, TypeScript and React.

Consulting Engineer (Hybrid), June 2022 – Present

Keyence

- Working within a highly collaborative environment assisting sales engineers and clients with technical detail necessary to accurately specify computer vision systems for automation considering client application requirements.
- Consulting with various system integrators and end users to ensure the success of applications.
- Responsible for programming proprietary software to meet client requirements.
- Working within single threaded workflow applying and optimising algorithms for image analysis. System architecture for I/O interfaces and asynchronous operation.
- Planning and executing seminars and webinars for client training and development.
- Applying bespoke strategies for client interaction in conjunction with SEs and management. Assess and scope our support training structure to enhance technical understanding, competency and repeat business value.
- Managing internal projects to enhance technical information sharing including presentation and dissemination.
- Hybrid working with a focus on site visits across mainland UK, at home software engineering where appropriate.
- Clients include; Jaguar Land Rover, Glaxo Smith Klyne, Dyson, Red Bull, Servomex, Amazon and various system integrators including: Harwin, Horizon Instruments and Huxley Bertram.

Single Consultation (Freelance), Nov 2019 – Jan 2020

Lazer Madness (Entertainment Center)

- Maintenance and upgrades of desktop PC's, all in one windows machines, Linux (Ubuntu Server) database, DAC and Amp configuration. Intercommunication system installation.
- Software maintenance and upgrades, bloatware and adware removal, firewall configuration including virus and malware security.
- Ongoing mobile support with applications and general questions.
- Documentation produced for all operations performed, applications installed and replaced hardware. Detailed technical explanations produced for new systems.

Skills

- **Mathematical:** Functions; Methods; Data aggregation; Boolean logic & Discrete Mathematics.
- **English Language:** Thematic content & context; Syntax & semantics.
- **English Literature:** Comprehension of various historical interpretations and modern language. Ambiguity avoidance.
- **Autonomous Self-Management and Intuition:** Competence in self-management organization and scheduling. Information collation revision (independent learning)
- **Software Development (Intermediate – Advanced):** C Variations, Python, various Object Orientated & Functional languages. Single threaded and multithreaded

systems. HTML & CSS (Java in-line). Databases; SQL variations. Linux distributions (Predominately Ubuntu Server & Kali). Experience with C-make and Catkin. **Game Development:** Unity scripting, including VR. **Web Development:** HTML5, CSS3, XML, JavaScript, JSON. **Internet of Things:** Esp8266, Arduino, MQTT, UDP.

- **Robotics:** Omni-directional forward and inverse Kinematics, actuation, and sensing
- **Electronics & Computer Hardware:** Stepper & Servo motors. Standardized PC hardware. Experience with various proprietary hardware products.
- **Analytics and Reviews:** Data analytics; trends, correlation and causation analysis.
- **Communication and Engagement:** Participation in group projects and activities in conjunction with senior staff and fellow peers including peer reviews and retrospectives.
- **Report Compilation:** Gathering relevant data from first and secondhand sources to achieve a project goal.
- **External Studies:** Studies and research conducted in spare time to supplement taught material, focus scoped to heterogeneous systems and overviews of various technology or according to job specification and business requirements.

Additional Information

- **Travel: Driving license holder** (7 years NCD, no points or convictions);
- **Familiarity with Rail** (C2C, SWR, London Tube).
- **Salary Requirements:** 60K – 80K preferable, seeking a fulfilling role rather than monetary gain.

Interests and Recreational Activities

- **Motorcycle enthusiast:** Involvement within the University of Portsmouth Motorcycle society, group rides and road safety management.
- **Gaming & audio-visual entertainment:** Avid video game player, with a deeper interest in design and creation within audio visual entertainment content and the application of technology within this sector.
- **Photography:** personal and semi-professional photography both true to life and artistic interpretations.
- **Watersport & extreme sports:** Fresh water wakeboarding, saltwater jet skiing, leisurely swimming for relaxation and exercise.
- **Tech industry affairs:** Enjoyment in keeping up to date with new developments and news within the Tech industry.

References are available on request.